Economics - Structural Correspondence with the 5-Stage Model

2026-03-21

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Economics

Structural correspondence survey with the 5-stage model (Ba / Wave / En / Uzu / Taba)

Survey Overview

- **Survey target**: 10 major theories in economics
- **Research question**: Do the various theories of economics correspond structurally to the 5-stage model?
- **Results**: Strong correspondence in 7 cases, partial correspondence in 3 cases

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Structural Correspondence Diagram

Overview of the 5-Stage Model

Stage	Definition
Ba (Field)	An undifferentiated state. The initial condition in which no direction or structure has yet been established
Wave	The exploratory stage in which multiple directions diverge and compete
En (Edge)	A state of tension in which opposing elements coexist without converging on either side. The place where they meet at the boundary, influence each other, and relationships emerge
Uzu (Vortex)	The stage in which a new coherence (order) arises spontaneously from within the tension
Taba (Bundle)	The stage in which the form is fixed and stabilizes as a reusable structure

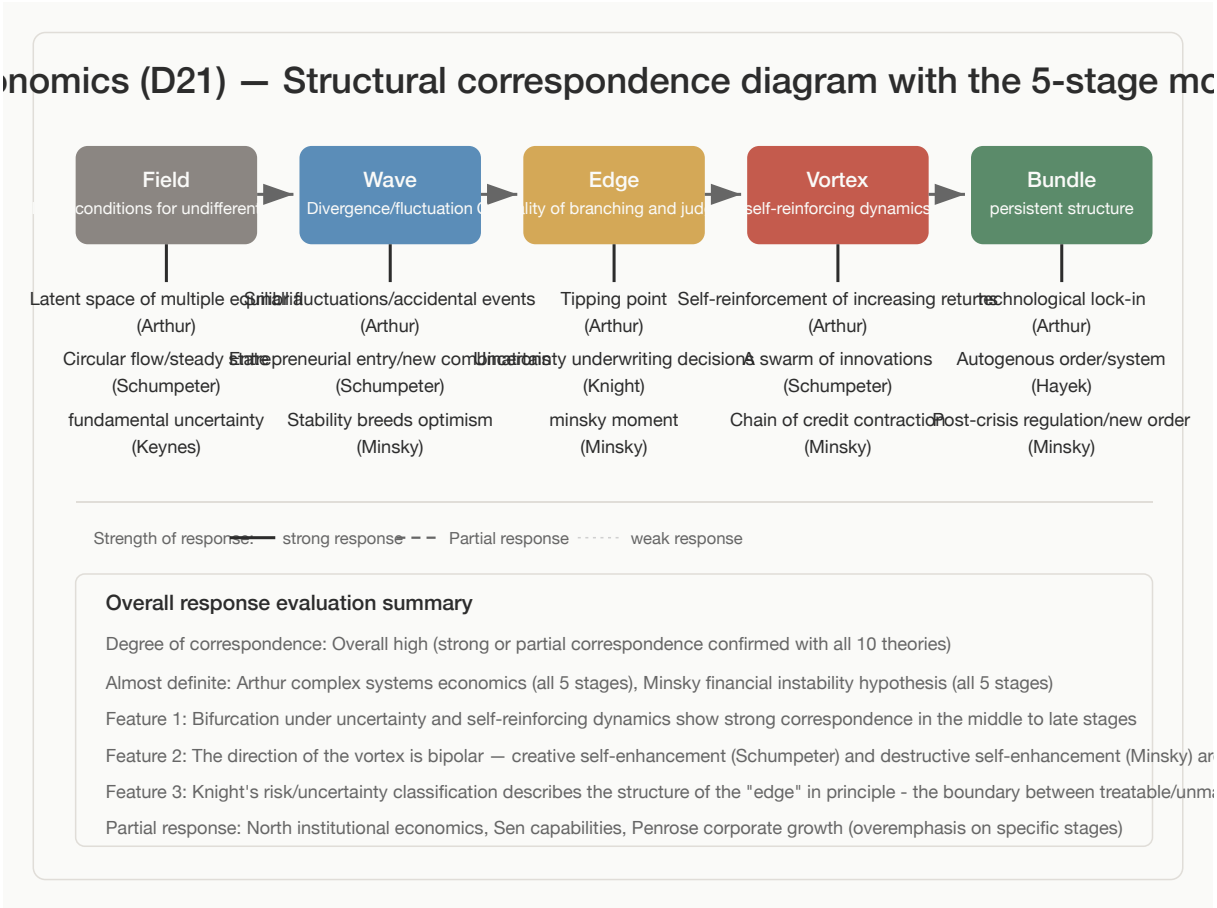

Figure 1: Structural correspondence diagram

Overview of Structural Correspondences

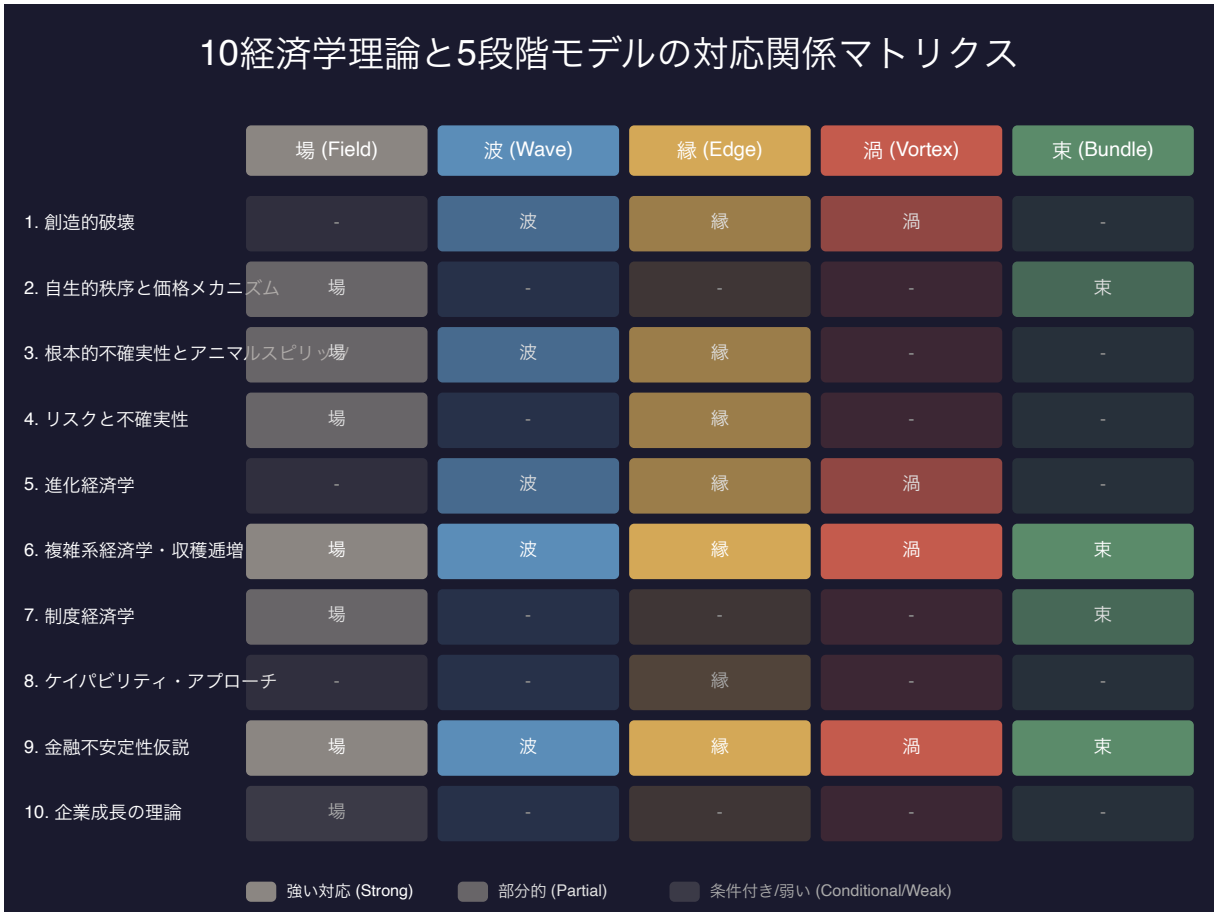

Figure 2: Domain — Theory x 5-Stage Correspondence Matrix

#	Theory/Concept	Proponent	Particularly Strong Correspondence Stage	Assessment
1	Creative Destruction	Schumpeter (1912, 1942)	Wave → Uzu	Strong correspondence
2	Spontaneous Order and the Price Mechanism	Hayek (1945, 1973)	Ba → Taba	Strong correspondence
3	Radical Uncertainty and Animal Spirits	Keynes (1936, 1937)	Ba → En	Strong correspondence
4	Risk and Uncertainty	Knight (1921)	En	Strong correspondence
5	Evolutionary Economics	Nelson & Winter (1982)	Wave → Uzu	Strong correspondence
6	Complexity Economics and Increasing Returns	Arthur (1994, 2015)	All 5 stages	Strong correspondence
7	Institutional Economics	North (1990)	Ba → Taba	Partial correspondence

#	Theory/Concept	Proponent	Particularly Strong Correspondence Stage	Assessment
8	Capability Approach	Sen (1999)	En	Partial correspondence
9	Financial Instability Hypothesis	Minsky (1986)	All 5 stages	Strong correspondence
10	Theory of the Growth of the Firm	Penrose (1959)	Ba → Uzu	Partial correspondence

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Key Entry 1: Arthur's Complexity Economics and Increasing Returns

- W. Brian Arthur is an economist who reframed the economy as a "complex adaptive system." Against the diminishing returns and stable equilibria assumed by conventional equilibrium theory, he showed that under conditions of increasing returns, small accidental advantages self-reinforce, leading to "lock-in" in which a particular path becomes fixed.
- **As a matter of fact**: Arthur argued that under conditions of increasing returns, multiple equilibria are possible, and which equilibrium is realized depends on small initial events (*Increasing Returns and Path Dependence in the Economy*, 1994). Central to this process is the "tipping point" — the critical juncture at which a slight difference in adoption, amplified through positive feedback, converges on a single path. In later work, he argues that the economy is inherently non-equilibrial and that macro-patterns emerge from the interactions of heterogeneous agents (*Complexity and the Economy*, 2015).
- **As an interpretation of structure**: Here we read a process structure in which multiple stable states compete and the path is determined at a critical point, then self-reinforcingly fixed. The level of similarity is structural. Rather than the closeness of labels, attention is given to the relational arrangement of "undecided → critical → self-reinforcing → fixed."
- **As an interpretive reading**: This structure corresponds clearly to the 5-stage model. The latent space in which multiple equilibria are possible is Ba; small fluctuations and accidental events are Wave; the tipping point — the critical juncture where multiple paths coexist and neither has yet been confirmed — is En; the self-reinforcing dynamics of increasing returns are Uzu; and the establishment of technological lock-in or a dominant design is Taba. Historical cases such as VHS vs. Betamax and the QWERTY layout are also instructive examples in which this 5-stage process can be traced concretely.
- There is a further point worth noting. In Arthur's theory, this structure repeats isomorphically at the level of individual technology, industry, and the economy as a whole. Tipping points occur in the adoption competition of individual technologies, in industry-wide standardization competition, and in transitions of economic regimes — the same structure appears in all cases.

Key Entry 2: Minsky's Financial Instability Hypothesis

- Hyman Minsky is an economist who proposed the paradoxical thesis that the stability of the financial system itself sows the seeds of destabilization. He was widely reassessed in the wake of the 2008 global financial crisis.
- **As a matter of fact**: Minsky argued that during periods of stability, economic actors expand risk-taking and financial structures become progressively fragile (*Stabilizing an Unstable Economy*, 1986). Financial postures change in three phases: first, hedge finance — a sound state in which principal and interest payments are covered by operating income; second, speculative finance — a state in which interest payments are possible but refinancing is needed for principal repayment; third, Ponzi finance — a state in which even interest payments depend on rising asset prices. The optimism of stable periods accelerates this transition, eventually reaching a turning point at which credit expansion rapidly reverses into deleveraging. This turning point later came to be called the "Minsky Moment" (named by McCulley, around 1998).

- **As an interpretation of structure**: Here we read a process structure in which stability generates instability from within, and once a critical point is crossed, a self-reinforcing chain of collapse unfolds. The level of similarity is structural, with particular attention to the relational arrangement of "stability → endogenous fragility → critical point → chain reaction → new order."
- **As an interpretive reading**: This structure corresponds to the 5-stage model. The financial order of the stable period is Ba; the optimism and expanding risk-taking that stability generates are Wave; the Minsky Moment at which Ponzi finance collapses is En; the self-reinforcing credit contraction through cascading asset sales is Uzu; and the post-crisis tightening of regulations and the formation of a new financial order are Taba.
- What is distinctive about this theory in economics is the point that "Ba endogenously generates its own destruction." In Schumpeter's creative destruction, entrepreneurs bring new combinations from outside; in Minsky, the stable system itself is the origin of instability. A further important point is that the Uzu here is "destructive." Whereas Nelson & Winter's Uzu is the consolidation of new routines, and Schumpeter's Uzu is the creative self-reinforcement of clustering innovation, Minsky's Uzu is the destructive self-reinforcement of cascading credit contraction. The structure (self-reinforcing dynamics) is isomorphic, but the direction differs.

Key Entry 3: Knight's Risk and Uncertainty

- Frank Knight is an economist who drew a strict distinction between "risk" and "uncertainty." This distinction became the foundation of research on uncertainty in economics.
- **As a matter of fact**: Knight defined situations in which a probability distribution is known as "risk," and situations in which probabilistic calculation itself is impossible as "uncertainty" (*Risk, Uncertainty, and Profit*, 1921). The source of profit lies in unmeasurable uncertainty; under perfect information, profit converges to zero. The essential function of the entrepreneur is to "bear" this uncertainty, and judgment is the core of entrepreneurship. Because uncertainty can be neither insured nor diversified, it must institutionally be "borne by someone."
- **As an interpretation of structure**: Here we read a branching structure in which, under unmeasurable conditions, a fork arises between "bearing" and "holding back," and this judgment determines economic outcomes. The level of similarity is mechanism, with particular attention to the branching function performed by the act of "judgment."
- **As an interpretive reading**: The sea of uncertainty — the domain where probabilistic calculation cannot be applied — corresponds to Ba. The phase in which uncertainty becomes manifest and existing predictive models become inadequate is Wave. And the entrepreneurial judgment — the fork between "bearing" and "holding back" in an unmeasurable situation — corresponds to En. It is a characteristic of this theory that the description of En is particularly clear. The generation of profit is feedback (Uzu) as the result of bearing uncertainty, and ex-post verification through market selection is Taba.
- Knight's "risk/uncertainty" distinction provides the most fundamental description of the branching condition in economics. Risk can be handled by probabilistic calculation (it can be discarded); uncertainty cannot be handled by calculation, and someone must either bear it or hold it. This distinction fundamentally shows the structure of En — the boundary between manageable and unmanageable variation.

Cross-Cutting Patterns

- Four patterns recur repeatedly across economics
- **First, "uncertainty" defines the structure of En
- **Second, "self-reinforcement" demonstrates diverse types of Uzu
- **Third, Ba is not "empty" but "conditional"

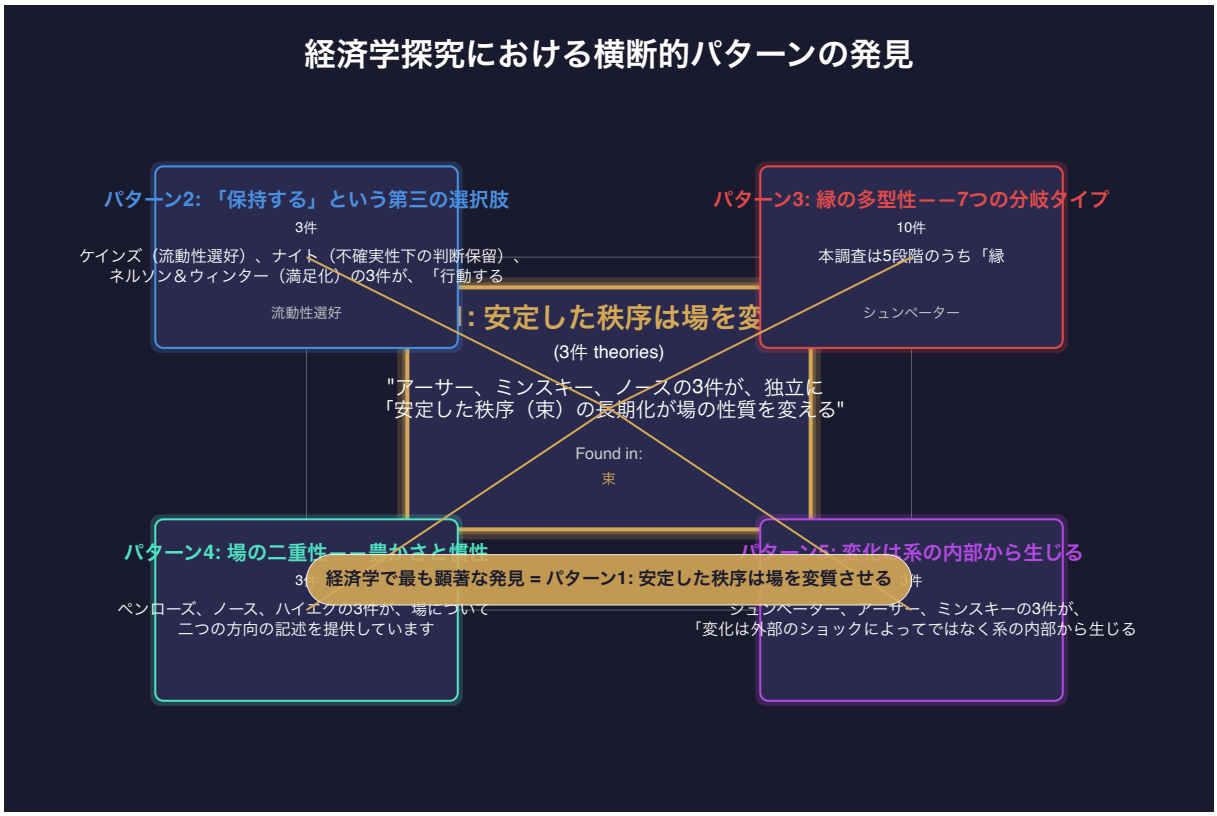

Figure 3: Domain — Cross-Cutting Patterns Diagram

Unresolved Questions

- The relationship between the "destructive Uzu" in Minsky's Financial Instability Hypothesis and the 5-stage model still requires clarification. The question remains whether Uzu must always be creative, or whether destructive self-reinforcement also falls within the scope of Uzu. However, if we adopt the view that post-crisis regulatory restructuring generates a new Taba, then destruction and creation can be read as different phases of the same cycle.
- Sen's Capability Approach is a framework for "evaluation" and not a description of a "generative process." The existence of a structural correspondence with the 5 stages and the alignment of the theory's problematic are separate issues. How to handle this difference is a matter for future consideration.
- All 10 theories in this survey belong to the lineage of "anti-equilibrium." Not including equilibrium theories (Walras, Arrow-Debreu) means that theories of the type that process fluctuations and recover to equilibrium were intentionally excluded. The influence of this selection judgment on the survey results needs to be recognized.
- In Hayek's spontaneous order and Arthur's tipping points, En is crossed without "anyone's conscious choice." If the 5-stage model is limited to conscious creation, these theories fall outside its scope; but if it also includes the generation of order without consciousness, they fall within it. This question has not yet been resolved.

Conclusion

- This survey confirmed that economics is a domain with overall high structural similarity to the 5-stage model
- As a finding close to established, economics is one of the domains that describes most fundamentally the structure of En mediated by "uncertainty"
- As compelling findings: the multi-directionality of Uzu — the same self-reinforcing structure acting both creatively and destructively; the conditionality of Ba — Ba is not empty but a precondition

filled with possibility; the repetition across scales — the same structure repeating at the level of technology, industry, and the economy as a whole

- Remaining at the hypothesis stage are: the correspondence with Sen's Capability Approach (the difference between evaluation and generation is not yet sorted out), the influence of the intentional exclusion of equilibrium theories on the results, and the positioning of "En without consciousness"